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Software engineering and information systems (4 years)

General information

Undergraduate studies in the according study program should provide for superb engineering staff capable of satisfying the ever-increasing demand for software engineers, especially programmers. Students enrolled in this study program acquire knowledge enabling them great professional flexibility and wide choice of employment opportunities wherever there is a need of analysis, specification, design, planning, implementation and programming, project management and maintaining software and information systems. Bachelor of Engineering of this profile can build his own career in software industry, in public and service sector, but especially in software development companies, real-time information systems, mobile and hybrid applications, internet and cloud software, advanced user interfaces, solutions related to design and implementation of software systems, etc.

The title obtained after completion of the study program

Bachelor of Science in Software Engineering and Information Technologies (240 ECTS)

Study program

In the curriculum, it is planned to have 29 compulsory subjects and 12 optional subjects. All optional subjects are divided into three difficulty groups (H1, H2 and H3). The weight is shown in the item code. Subjects containing L1 are subjects of weight group H1. Subjects containing L2 are subjects of weight group H2. Subjects that contain L3 in the code are subjects of weight group H3. The rules for optional subjects according to the level are defined as:

- Number of subjects from H1 - the total number of credits of selected elective subjects from group H1 must not exceed 6 credits.
- Number of H2 subjects - the total number of credits of selected elective subjects from the H2 group must not exceed 36 credits.
- Number of H3 subjects – there is no limit to the number of selected subjects from the H3 group.

The student can choose more courses from group H1 and group H2 than the number provided, but in that case the credits earned for these courses do not count towards the total number of credits needed to obtain the diploma. The student can receive a maximum of 6 ECTS credits from extracurricular activities approved by the teaching and scientific council of the faculty. Passed subjects and earned credits will be part of the list of passed subjects in the certificate of passed subjects after graduation from the faculty.

Subjects

Semester: 1 Semester: 2 Semester: 3 Semester: 4 Semester: 5 Semester: 6 Semester: 7 Semester: 8

Semester: 1



Compulsory subjects

Code	Subject
F23L1W004	Sport and Health (/subject/F23L1W004/en)
F23L1W005	Business and Management (/subject/F23L1W005/en)
F23L1W007	Introduction to computer science (/subject/F23L1W007/en)
F23L1W018	Professional skills (/subject/F23L1W018/en)
F23L1W020	Structured programming (/subject/F23L1W020/en)
F23L2W002	Mathematics 1 (/subject/F23L2W002/en)

Semester: 2



Compulsory subjects	
Code	Subject
F23L1S003	Computer Architecture and Organization (/subject/F23L1S003/en)
F23L1S016	Object-oriented programming (/subject/F23L1S016/en)
F23L2S001	Mathematics 2 (/subject/F23L2S001/en)
F23L2S015	Object-oriented analysis and design (/subject/F23L2S015/en)
	Elective subject with code starting with F23L1S F23L1S

Semester: 3



Compulsory subjects	
Code	Subject
F23L2W001	Algorithms and data structures (/subject/F23L2W001/en)
F23L2W014	Computer Networks and Security (/subject/F23L2W014/en)
F23L3W001	Mathematics 3 (/subject/F23L3W001/en)
	Elective subject with code starting with F23L2W F23L2W
	Elective subject with code starting with F23L2W F23L2W

Semester: 4



Compulsory subjects	
Code	Subject
F23L2S002	Software requirements analysis (/subject/F23L2S002/en)

F23L2S017	Operating systems (/subject/F23L2S017/en)
F23L2S030	Artificial Intelligence (/subject/F23L2S030/en)
F23L3S100	business practice (/subject/F23L3S100/en)
	Elective subject with code starting with F23L2S F23L2S

Semester: 5



Compulsory subjects	
Code	Subject
F23L3W004	Databases (/subject/F23L3W004/en)
F23L3W008	Introduction to datascience (/subject/F23L3W008/en)
F23L3W009	Software design and architecture (/subject/F23L3W009/en)
F23L3W140	Advanced programming (/subject/F23L3W140/en)
	Elective subject with code starting with F23L2W или F23L3W F23L2W F23L3W

Semester: 6



Compulsory subjects	
Code	Subject
F23L3S010	Human-computer interaction design (/subject/F23L3S010/en)
F23L3S012	System Integration (/subject/F23L3S012/en)
F23L3S019	Software quality and testing (/subject/F23L3S019/en)
F23L3S138	Advanced databases (/subject/F23L3S138/en)
	Elective subject with code starting with F23L2S или F23L3S F23L2S F23L3S

Semester: 7



Compulsory subjects	
Code	Subject
F23L3W021	Team project (/subject/F23L3W021/en)
	Elective subject with code starting with F23L3W F23L3W
	Elective subject with code starting with F23L3W F23L3W

	Elective subject with code starting with F23L3W F23L3W
	Elective subject with code starting with F23L3W F23L3W

Semester: 8



Compulsory subjects	
Code	Subject
F23L3S022	ICT Projects Management (/subject/F23L3S022/en)
F23L3S028	Entrepreneurship (/subject/F23L3S028/en)
F23L3S168	Diploma thesis (/subject/F23L3S168/en)
	Elective subject with code starting with F23L2S или F23L3S F23L2S F23L3S
	Elective subject with code starting with F23L3S F23L3S

Elective subjects from group: F23L1S



Code	Предмет	Suggested semester	Dependencies
F23L1S052	E-learning (/subject/F23L1S052/en)	2	
F23L1S116	Computer components (/subject/F23L1S116/en)	2	
F23L1S120	Creative skills for problem solving (/subject/F23L1S120/en)	2	нема
F23L1S146	Fundamentals of Web design (/subject/F23L1S146/en)	2	
F23L2S066	F23L1S066 Cybersecurity fundamentals (/subject/F23L2S066/en)	2	нема

Elective subjects from group: F23L2S



Code	Предмет	Suggested semester	Dependencies
F23L1S026	F23L2S026 Marketing (/subject/F23L1S026/en)	4	Освоени најмалку 36 ЕКТС
F23L2S042	Electrical Circuits (/subject/F23L2S042/en)	4	Освоени најмалку 36 ЕКТС
F23L2S051	Computational thinking in education (/subject/F23L2S051/en)	4	Вовед во компјутерски науки
F23L2S061	Wireless and mobile systems (/subject/F23L2S061/en)	4	Освоени најмалку 36 ЕКТС